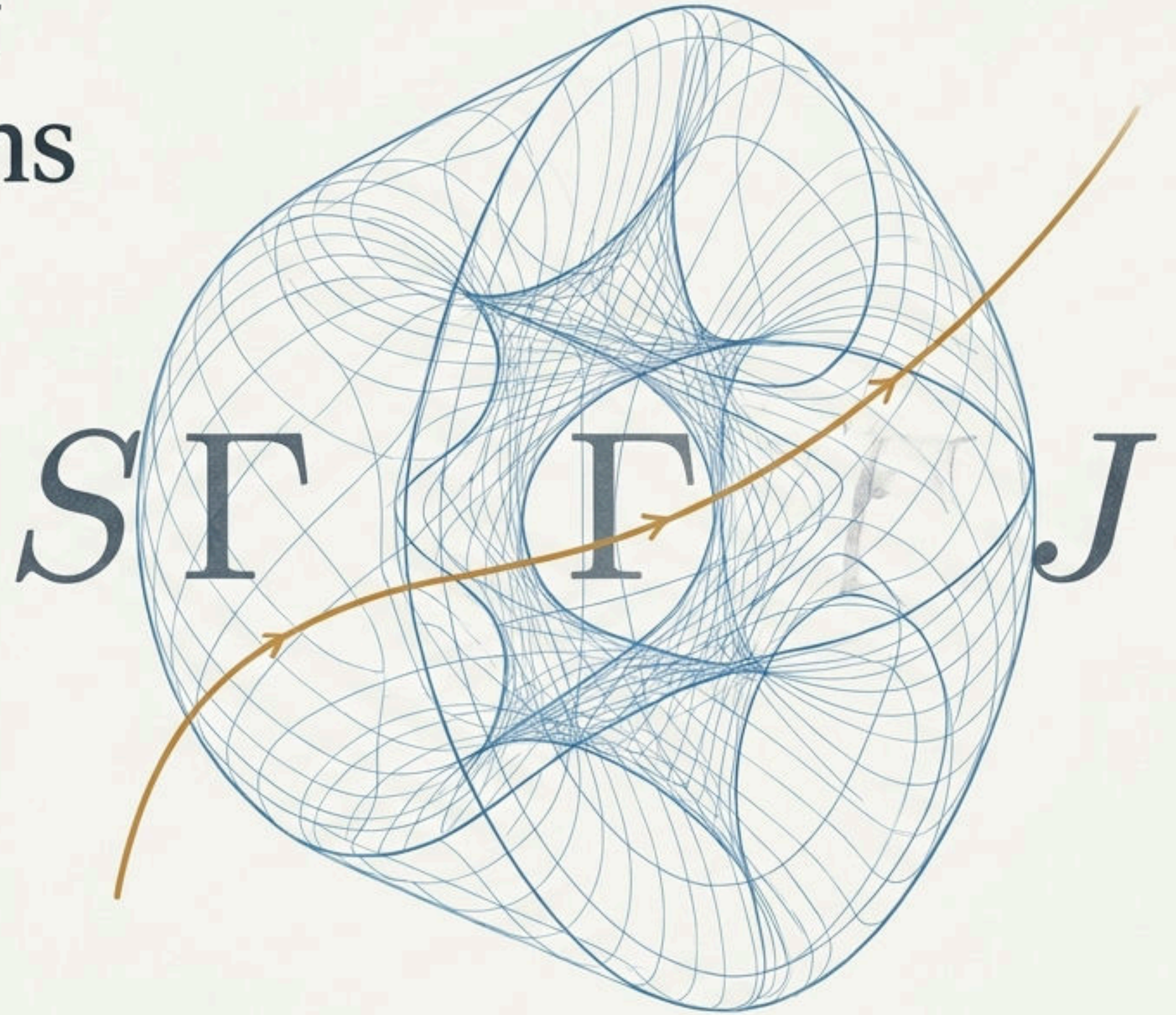


The Mathematics of Autonomous Systems

Replacing the metaphors of cognition, biology, and scientific discovery with a unified geometric formalism of admissible state spaces and recursive repair.



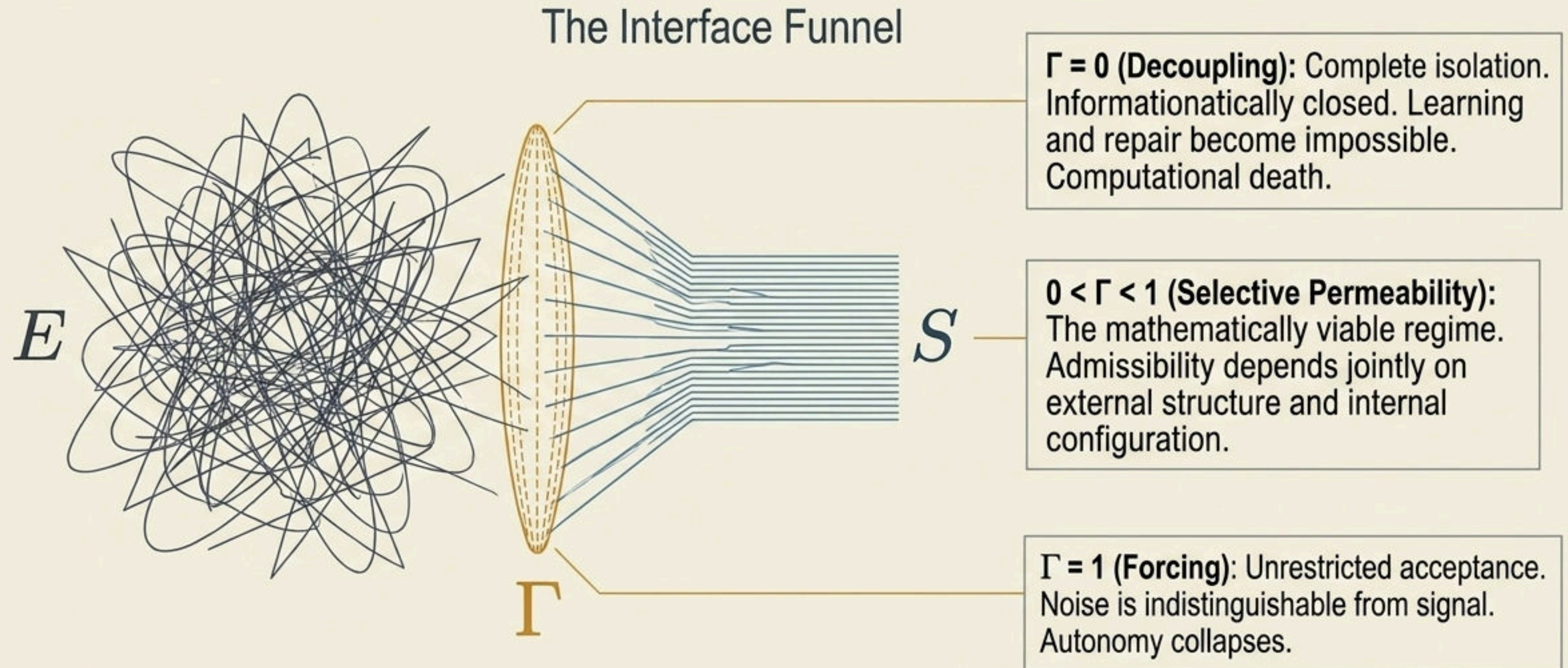
Based on the 2026 monograph by Flyxion.

Discarding informal metaphors for rigorous topological formalisms

The Informal Metaphor	The Mathematical Object
Sanctuary / Isolation	Autonomous Cognitive Domain
Silence / Walls	Admissible Interface / Entropy Minimization
Identity as Substance	Identity as Admissible Continuation
Direct Observation	Representation Defect
Solitary Genius	Distributed Repair Graph
Scientific Creation	Recursive Navigation of Theory Space

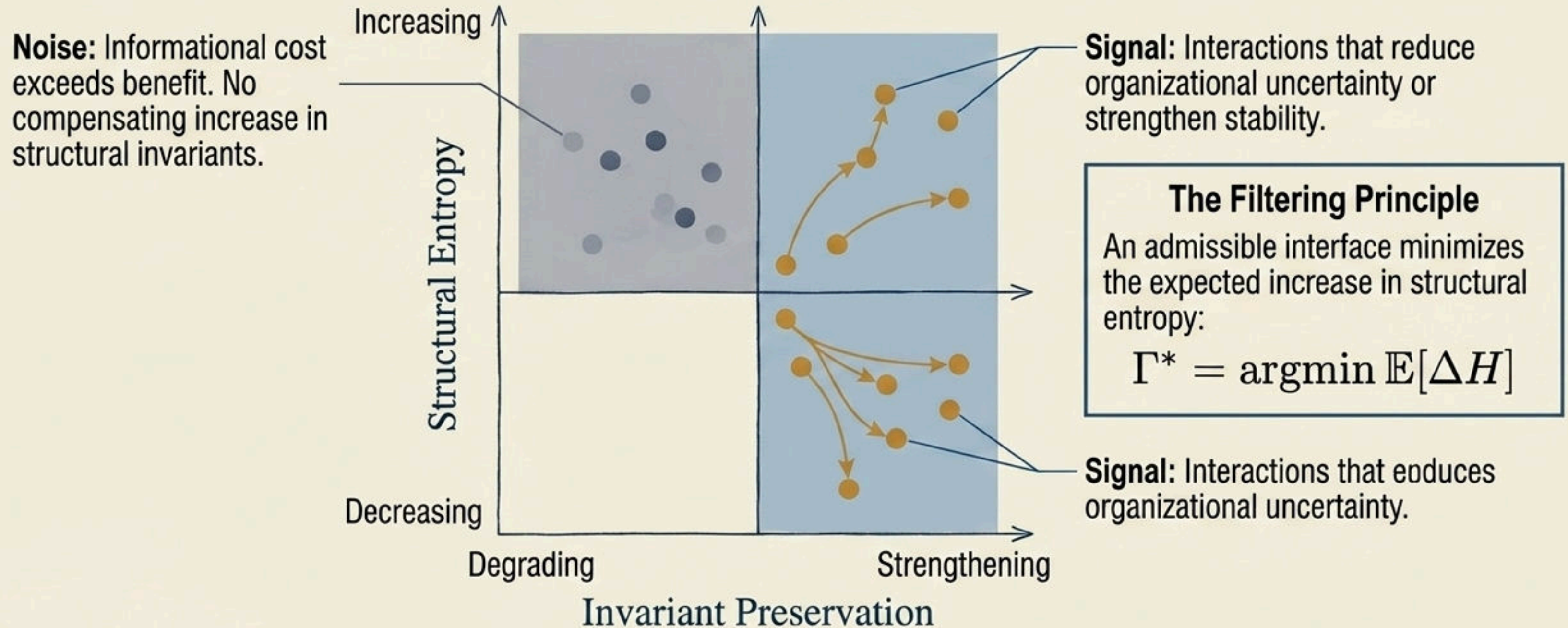
Key Insight: Autonomy is not isolation; it is invariant-preserving interaction.

The boundary is an active mathematical filter, not a passive wall



Key Insight: Every computationally active system must exchange information.
The goal is to constrain interaction, not eliminate it.

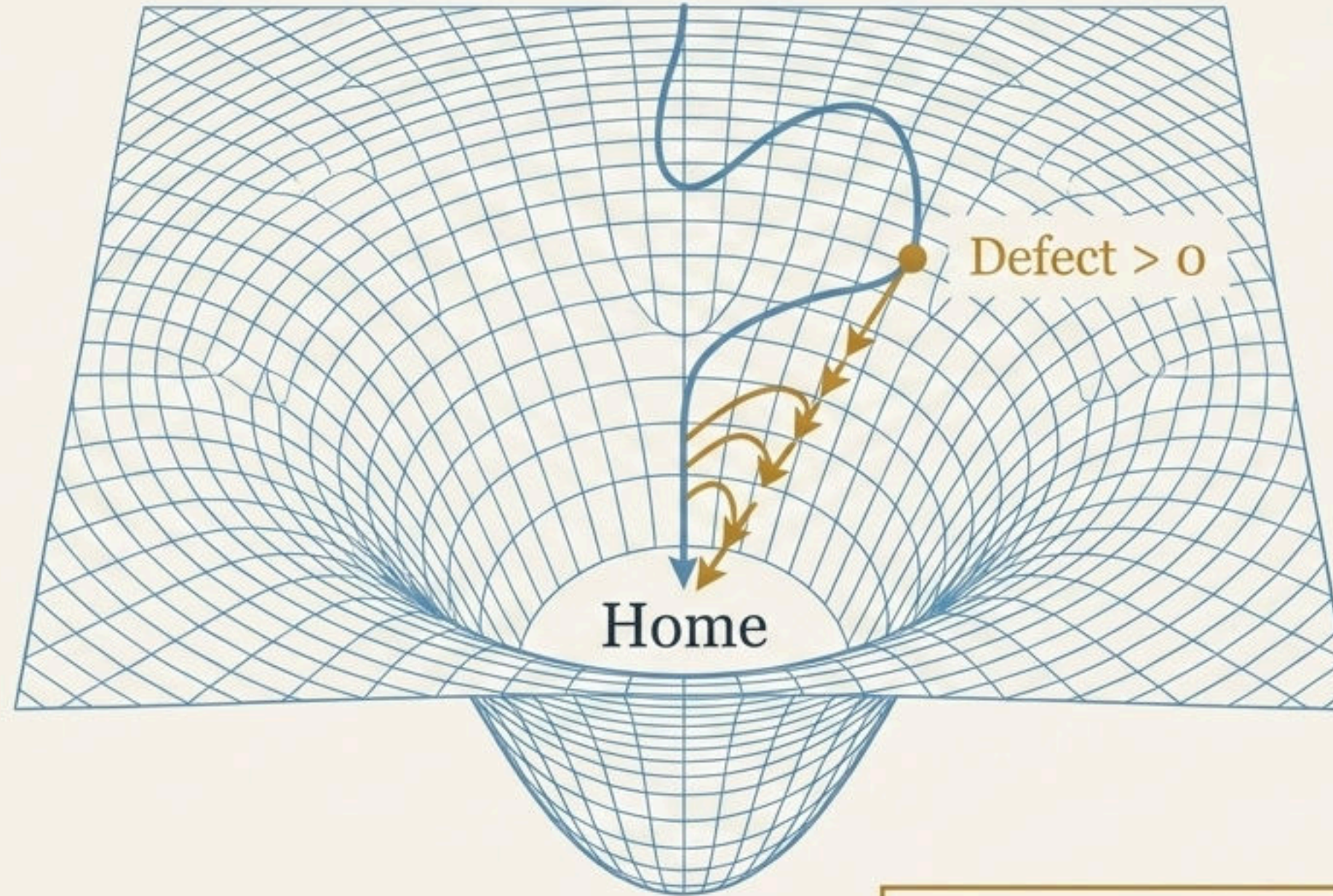
Environmental noise is relational, not an intrinsic absolute



Key Insight: An identical external event can be a catastrophic perturbation for one system and dynamically negligible for another.

Home is an invariant topological region, not a spatial location

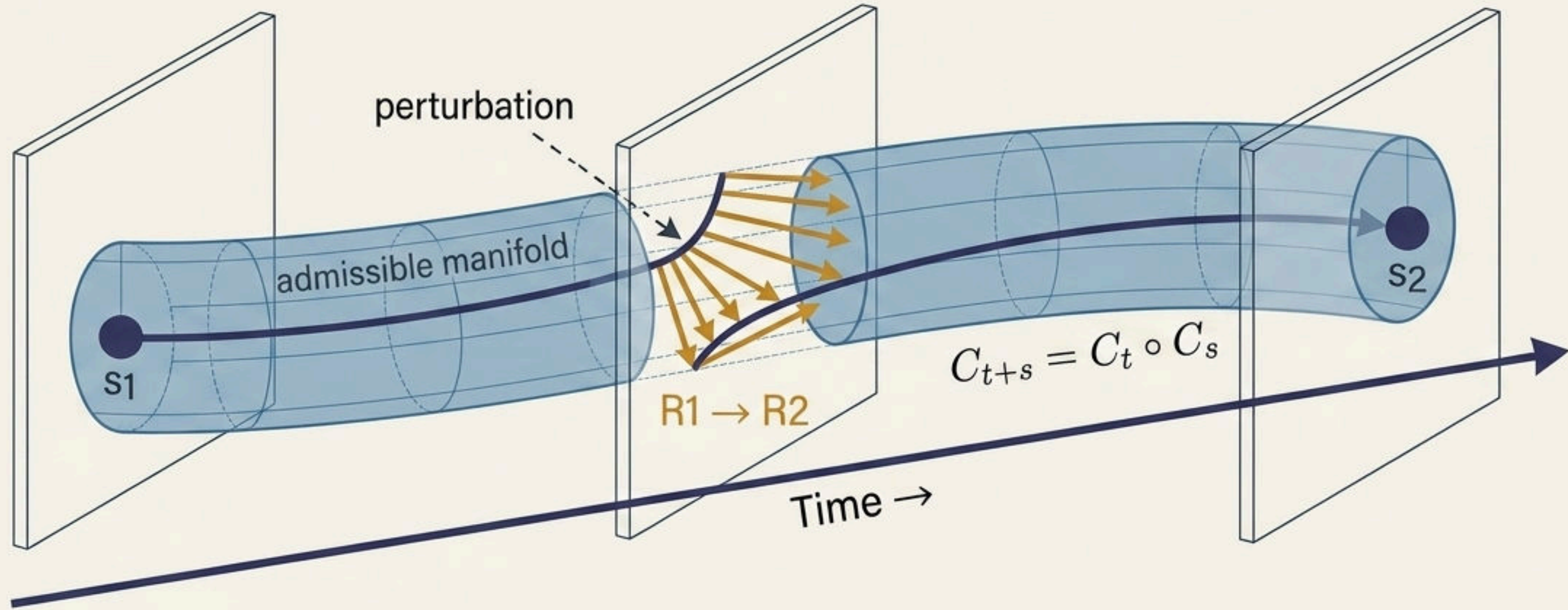
Defect: Measures geometric distance from invariant organization, not moral or psychological failure.



Recursive Repair:
Repair does not restore the past; it constructs an admissible future by repeatedly decreasing structural defect until limit defect = 0.

Key Insight: Home is not where a system begins. Home is where recursive repair converges.

Identity emerges from lawful continuation, not static substance

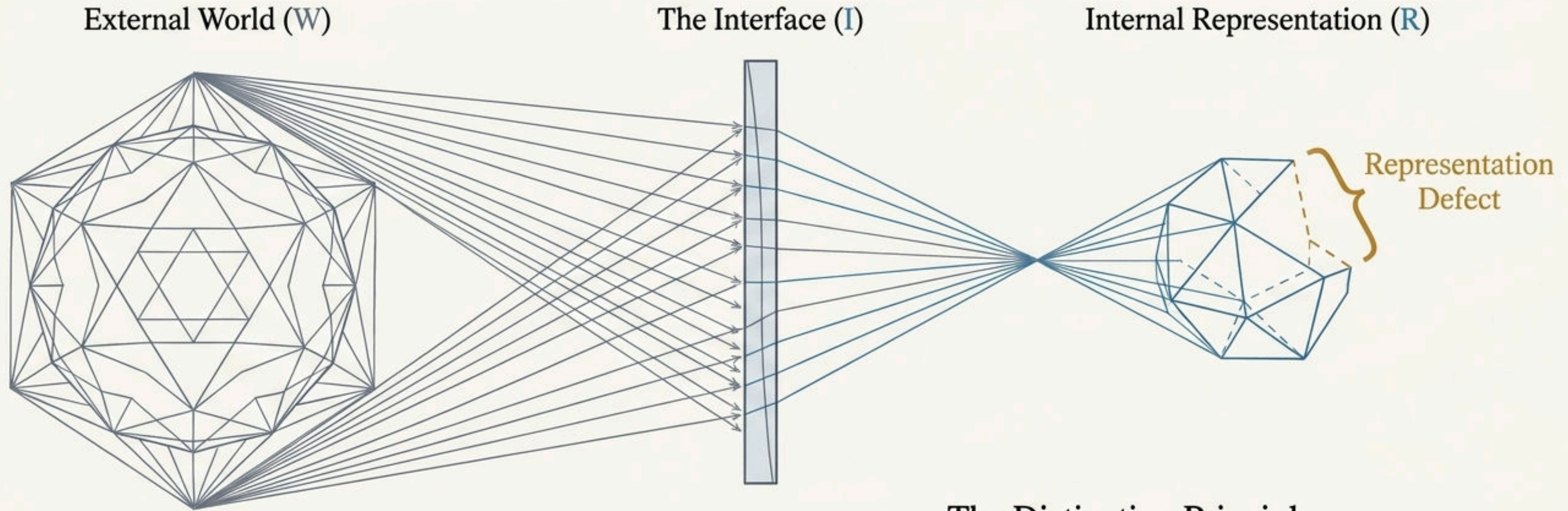


If identity required static substance, biological development, learning, and computation would all constitute the destruction of identity.

Identity equivalence is maintained as long as intermediate states remain within the admissible manifold.

Key Insight: Objects are transient realizations. Identity is the stable geometry of their admissible continuation.

Direct observation is mathematically impossible; information requires geometric distinction

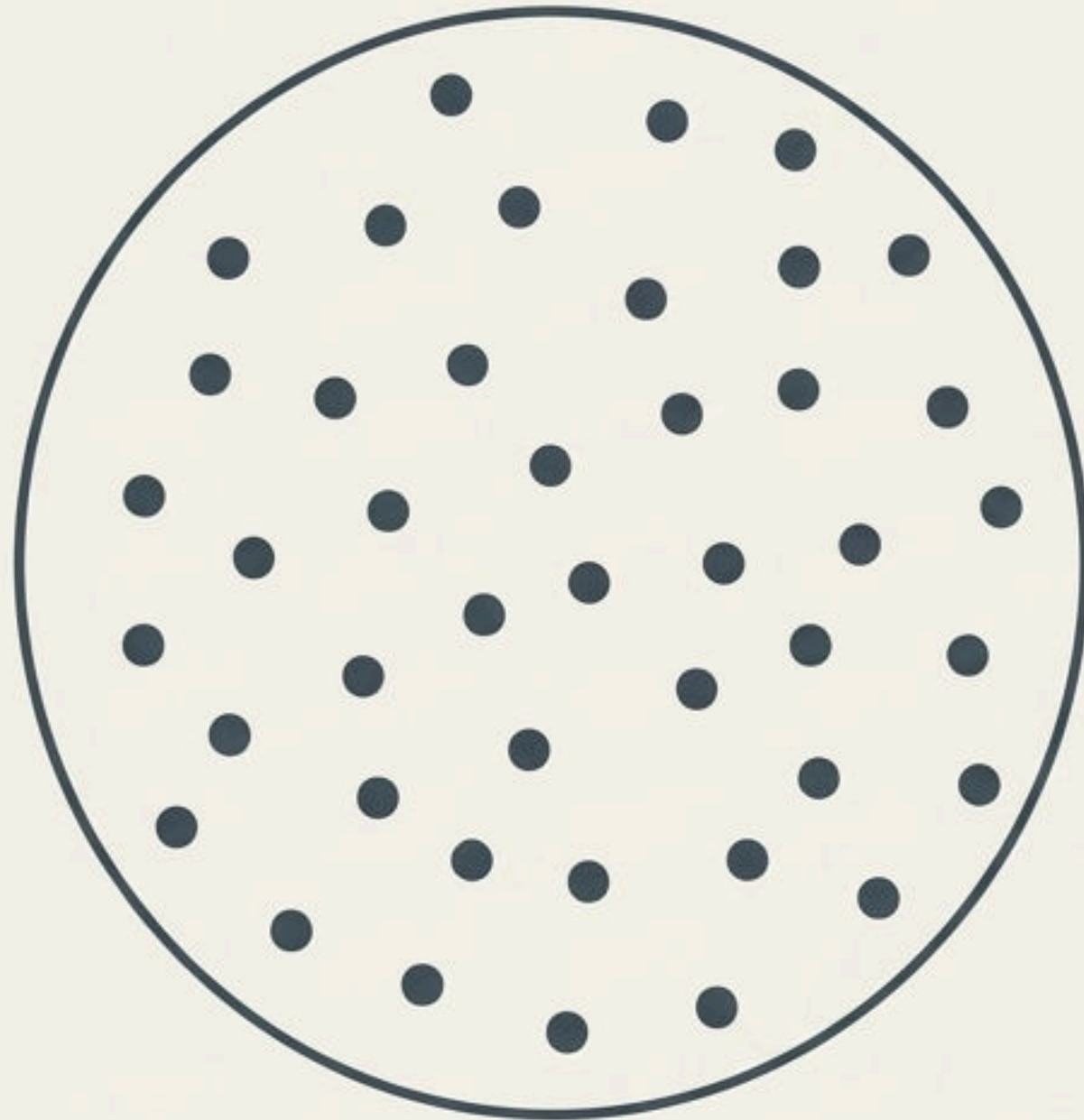


Key Insight: Information is downstream of distinction geometry. We reason over reconstructed equivalence classes, never reality itself.

The Distinction Principle

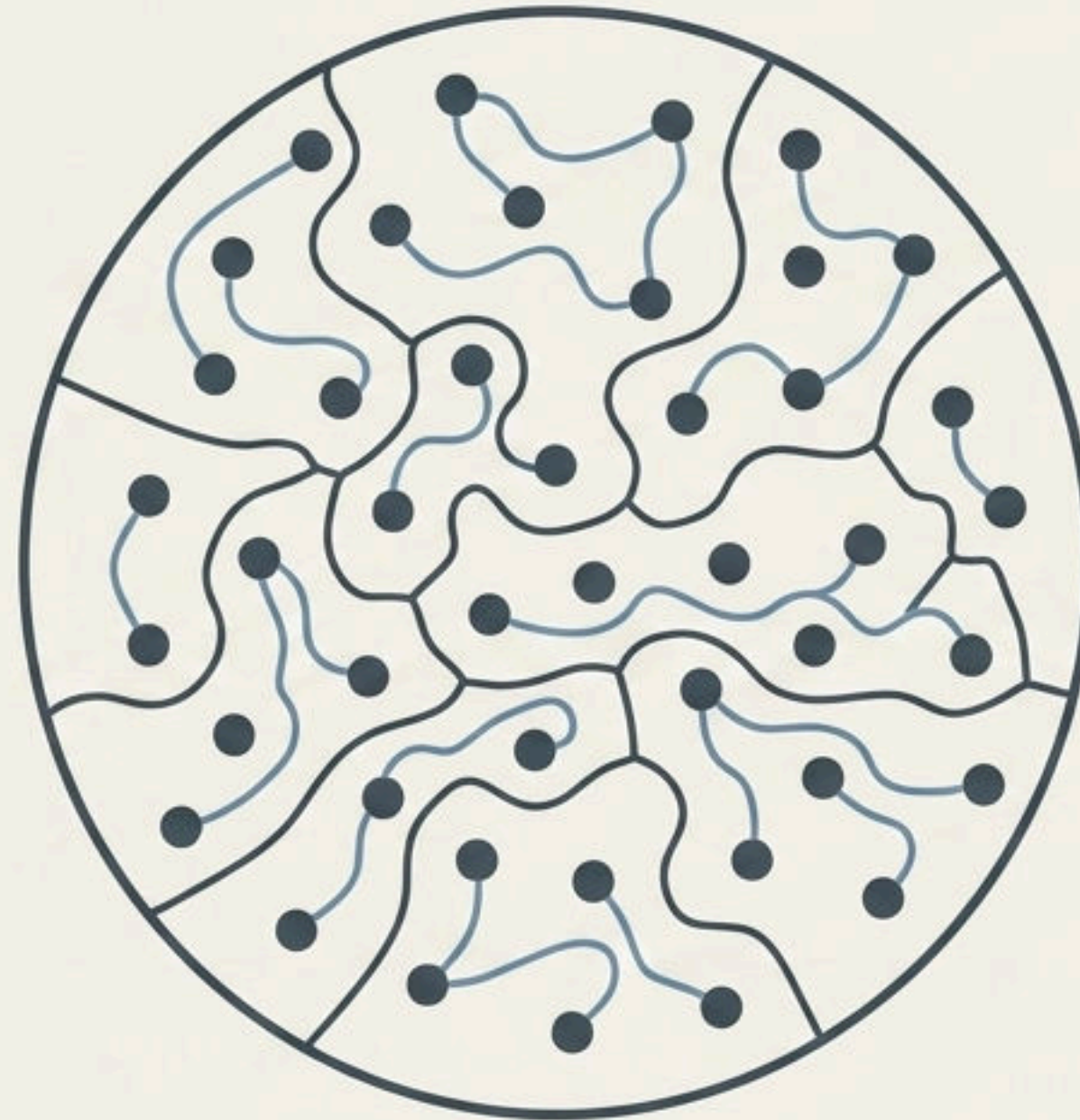
1. Information is not primitive. Before a probability distribution can exist, there must be **distinguishable** configurations.
2. Information = **Observation** $\circ$ **Distinction**.
3. If a finite interface collapses two realities into one equivalence class (**Observational Collapse**), no amount of internal computation can separate them.

Constraints precede motion: Success is about mapping navigable paths



Mere States

Implies any state can be achieved from any other.



Reachability Components

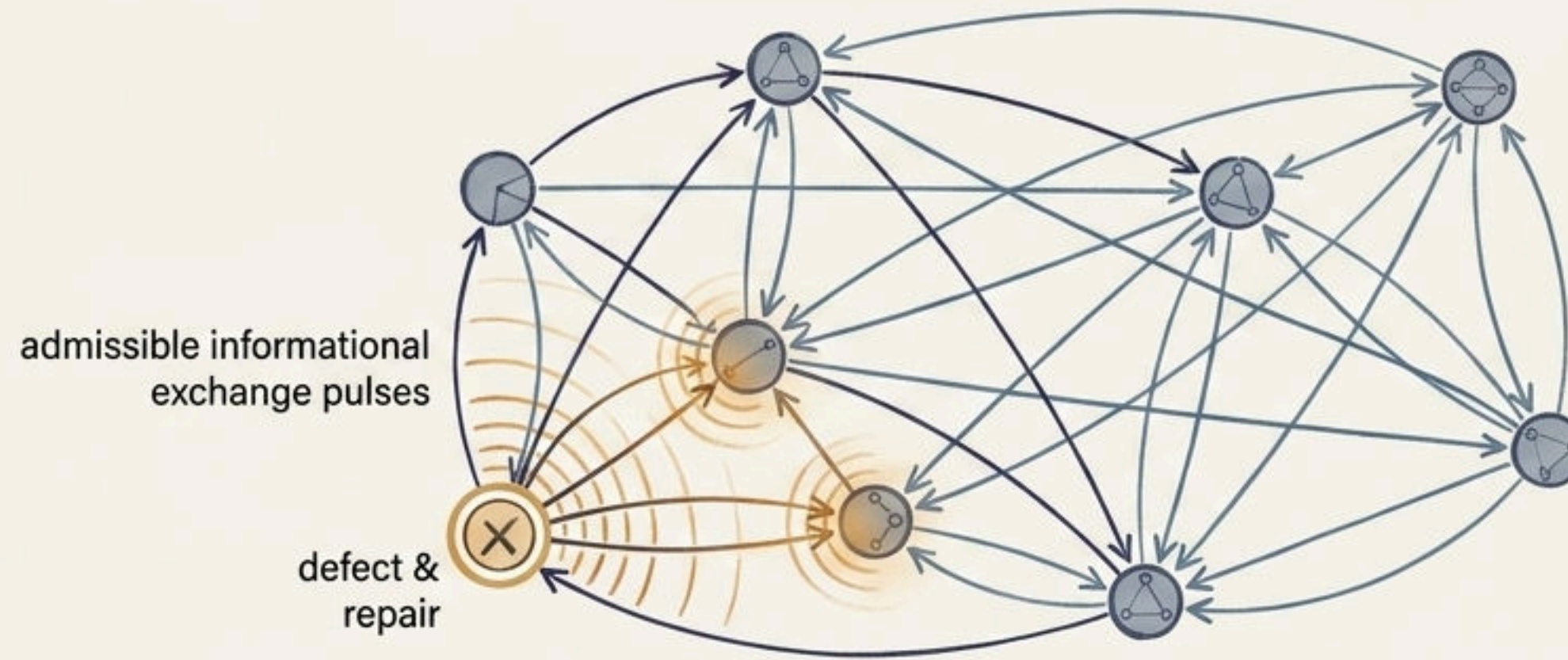
Boundary Geometry:
Arbitrarily small perturbations near a constraint boundary destroy admissibility.

Stable systems naturally evolve toward interior regions.

Constraint Energy:
Repair acts as geometric descent toward admissibility.

Key Insight: The existence of a destination does not imply the existence of an admissible path. Dynamics is entirely defined by constraint topology.

Distributed computation shatters the myth of the solitary genius



Autonomy is not
informational independence

No theorem proves itself; no
computation avoids inherited
infrastructure.

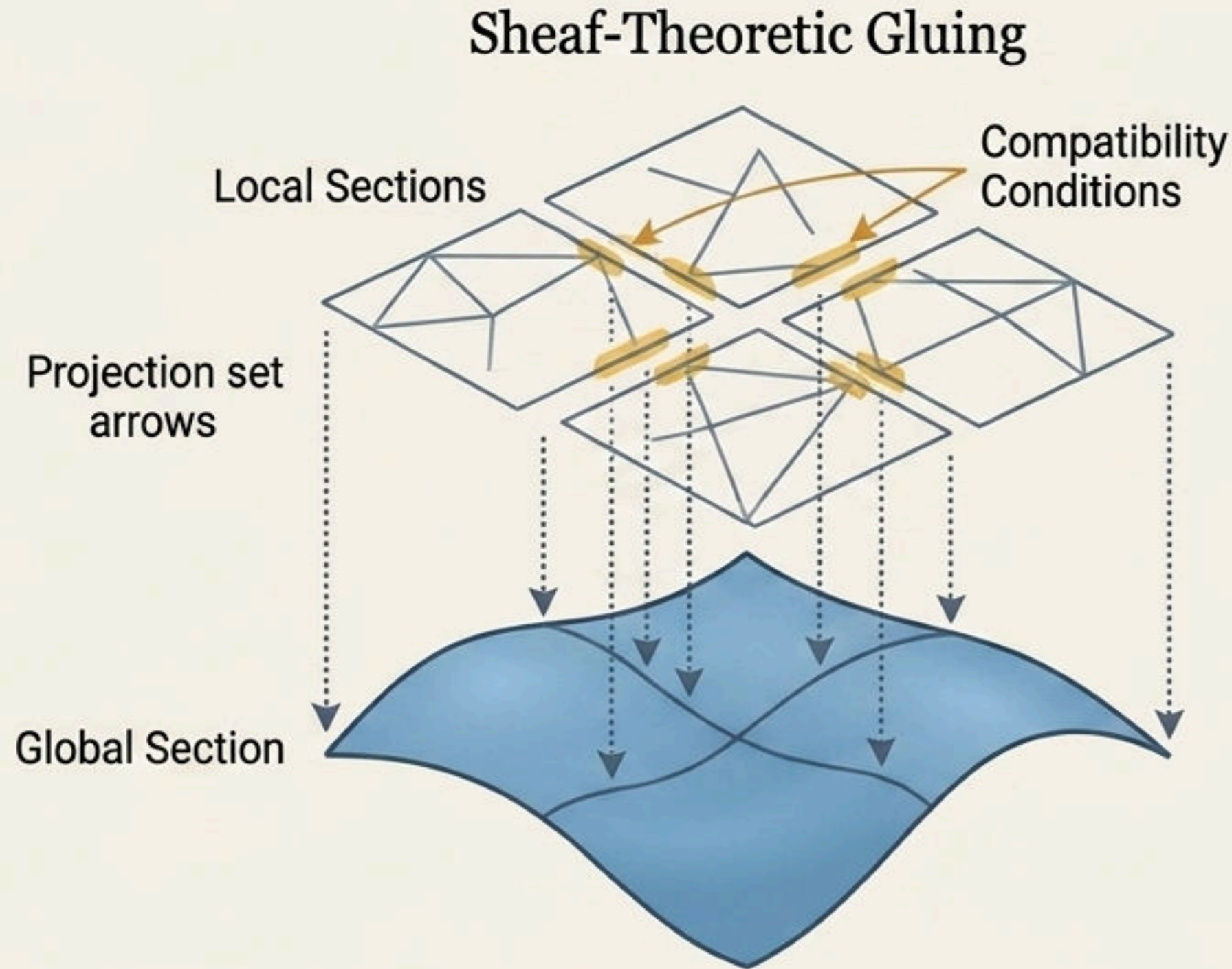
Indirect coupling allows
systems (e.g., Euclid, Newton)
to structurally influence modern
nodes without temporal
simultaneity.

Emergent Intelligence: The
global graph repairs what no
individual node can.

Knowledge propagates through
graph topology, not chronology.

Key Insight: The proper unit of study is the distributed repair graph.

Global truth emerges from the recursive gluing of overlapping local observations



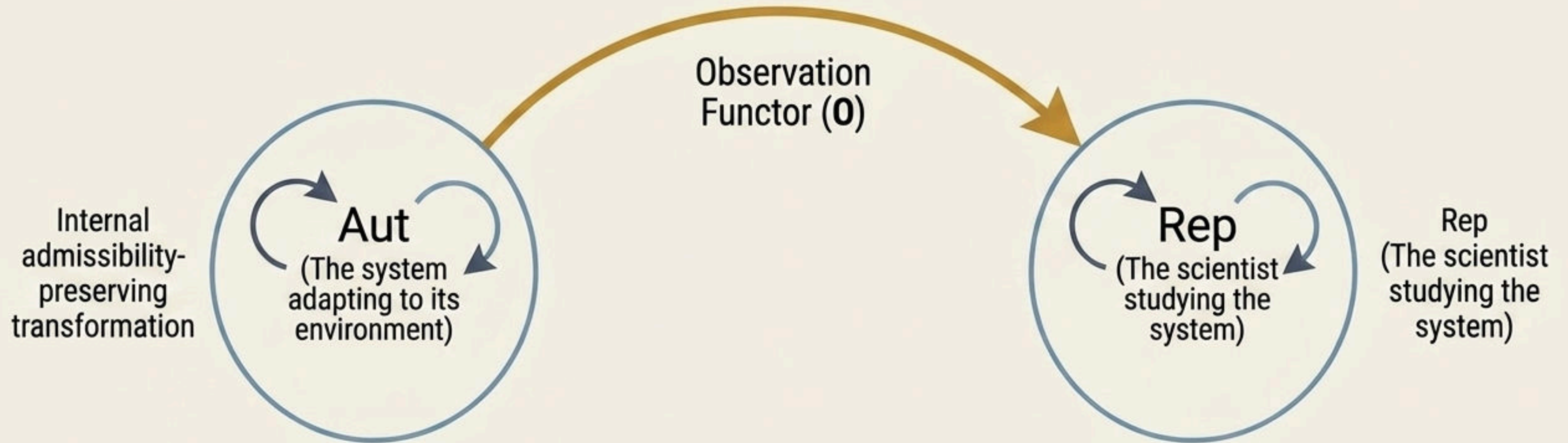
The Local Nature of Knowledge:
No single observer measures the entire system.

Reconstruction Defect:
Disagreement between neighboring observations on their overlaps.

Repair by Gluing:
Scientific collaboration acts by recursively minimizing overlap inconsistency, forcing local patches into global coherence.

Key Insight: Truth is not possessed by a privileged viewpoint; it is an emergent topological property of recursive compatibility.

The Functorial Principle: The observer and the observed are mathematically identical

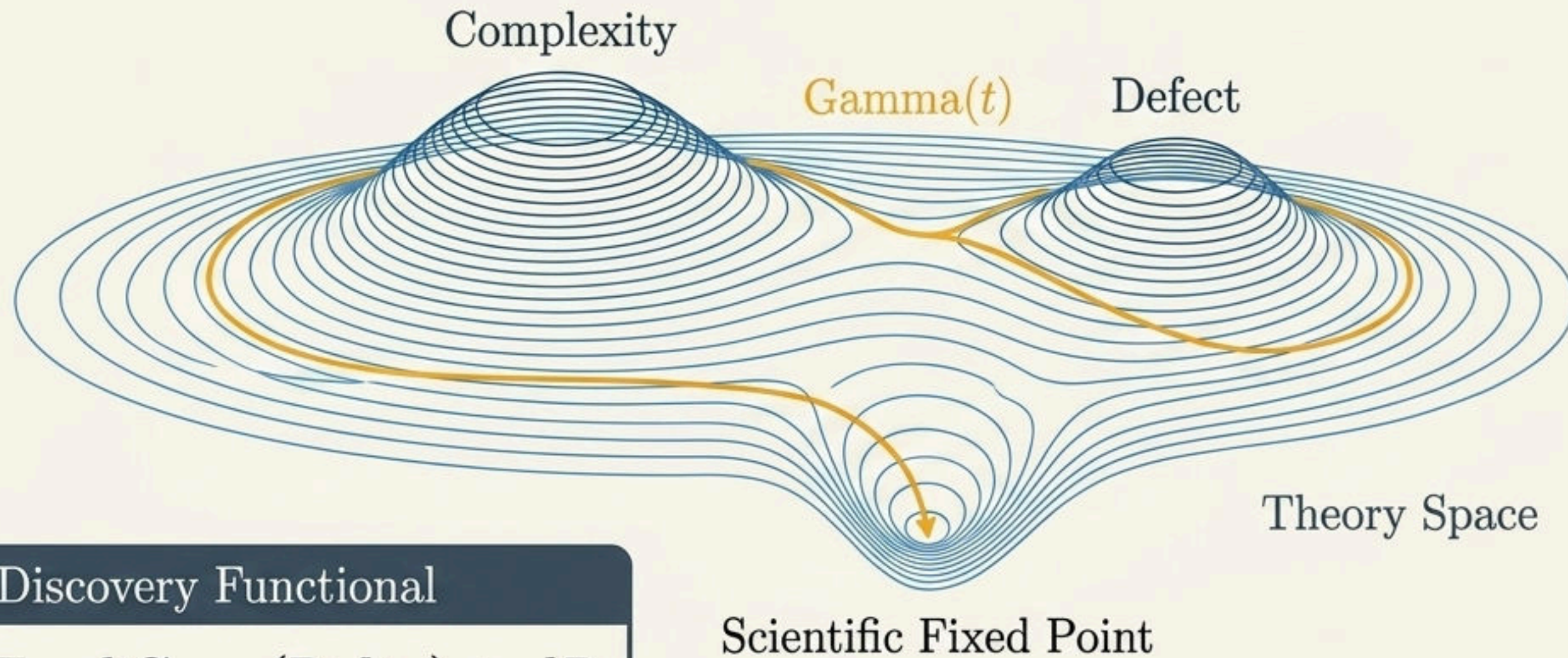


Scientific explanation does not catalogue isolated objects; it identifies functorially preserved structure.

Biology, software engineering, and scientific inquiry are all structurally equivalent instances of a single **endofunctor (Repair)** operating on autonomous state spaces.

Key Insight: The scientist interacting with a theory is mathematically identical to an organism interacting with its environment.

Scientific discovery is the recursive navigation of theory space



The Discovery Functional

$$D(T) = aE - bC - c(\text{Delta}) + dR$$

E = Explanatory power (Maximize)

C = Unnecessary complexity (Minimize)

Delta = Reconstruction defect (Minimize)

R = Admissible reachability (Maximize)

Key Insight: Paradigm shifts are not just sociological events; they are topological transitions that enlarge admissible regions in theory space.

The Principle of Recursive Scientific Realism

Scientific **progress does not culminate** in immutable certainty, nor is it merely unrestricted conceptual revision.

Reality is accessed **constructively**: a theory is accepted because it admits **indefinite, recursive repair** while preserving an ever-expanding family of mathematical and experimental invariants.

Concluding Insight: The objective of science is not static perfection. It is indefinite admissible continuation.